ONLY TWO POSSIBILITIES

Appending $\mathbf{b}$ as an extra column can raise the rank by at most 1. So comparing the two ranks leaves exactly two outcomes.

EXISTENCE

$A\mathbf{x} = \mathbf{b}$

split by $\text{rank}([A \mid \mathbf{b}])$ compared with $\text{rank}(A)$

✓

CASE 1

$\text{rank}([A \mid \mathbf{b}]) = \text{rank}(A)$

Consistent

$\mathbf{b}$ is a linear combination of the columns; appending it adds no new independent direction

✗

CASE 2

$\text{rank}([A \mid \mathbf{b}]) = \text{rank}(A) + 1$

Inconsistent

$\mathbf{b}$ is not in the column space; echelon form shows a row $[0 \ \cdots \ 0 \mid d]$ with $d \neq 0$

✓

TAKEAWAY

Existence is a question about $\mathbf{b}$ alone: is it already a linear combination of the columns of A?